

BUSFIN 8200: Problem Set 1

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1 Question 1

1.1 Part (a)

Assume $D_t > 0$. By definition, the gross return on equity is

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}. \quad (1)$$

Taking logs,

$$\begin{aligned} \log(R_{e,t+1}) &= \log\left(\frac{P_{t+1} + D_{t+1}}{P_t}\right) \\ &= \log\left(\frac{P_{t+1} + D_{t+1}}{P_{t+1}} \cdot P_{t+1}\right) - \log(P_t) \\ &= \log\left[\left(1 + \frac{D_{t+1}}{P_{t+1}}\right) P_{t+1}\right] - \log(P_t) \\ &= \log(P_{t+1}) + \log\left(1 + \frac{D_{t+1}}{P_{t+1}}\right) - \log(P_t). \end{aligned}$$

Define $r_{e,t+1} \equiv \log(R_{e,t+1})$, $p_t \equiv \log(P_t)$, and $d_t \equiv \log(D_t)$. Since $\frac{D_{t+1}}{P_{t+1}} = e^{\log(D_{t+1}/P_{t+1})} = e^{d_{t+1}-p_{t+1}}$, we can write

$$r_{e,t+1} = p_{t+1} + \log\left(1 + e^{d_{t+1}-p_{t+1}}\right) - p_t.$$

Define $f(x) \equiv \log(1 + e^x)$, where $x = d_{t+1} - p_{t+1}$. We approximate this function around the average log dividend-price ratio $\bar{d} - \bar{p}$ using a first-order Taylor approximation,

$$f(x) \approx f(\bar{d} - \bar{p}) + f'(\bar{d} - \bar{p})(x - \bar{d} - \bar{p}),$$

so that

$$f(d_{t+1} - p_{t+1}) \approx f(\bar{d} - \bar{p}) + f'(\bar{d} - \bar{p})[(d_{t+1} - p_{t+1}) - \bar{d} - \bar{p}].$$

Since $f(x) = \log(1 + e^x)$, we have $f'(x) = \frac{e^x}{1 + e^x}$, so

$$f'(\bar{d} - \bar{p}) = \frac{e^{\bar{d} - \bar{p}}}{1 + e^{\bar{d} - \bar{p}}}.$$

Define $\kappa \equiv \frac{1}{1 + e^{\bar{d} - \bar{p}}}$. Then $1 - \kappa = \frac{e^{\bar{d} - \bar{p}}}{1 + e^{\bar{d} - \bar{p}}}$, so $f'(\bar{d} - \bar{p}) = 1 - \kappa$.

Hence,

$$\begin{aligned} f(d_{t+1} - p_{t+1}) &\approx f(\overline{d-p}) + (1-\kappa) \left[(d_{t+1} - p_{t+1}) - \overline{d-p} \right] \\ &= f(\overline{d-p}) - (1-\kappa)\overline{d-p} + (1-\kappa)(d_{t+1} - p_{t+1}). \end{aligned}$$

Define $\kappa_0 \equiv f(\overline{d-p}) - (1-\kappa)\overline{d-p}$, a constant. Since $f(\overline{d-p}) = \log(1+e^{\overline{d-p}})$ and $\kappa = \frac{1}{1+e^{\overline{d-p}}}$, we have $1+e^{\overline{d-p}} = \frac{1}{\kappa}$, so

$$f(\overline{d-p}) = \log\left(\frac{1}{\kappa}\right) = -\log \kappa, \quad e^{\overline{d-p}} = \frac{1}{\kappa} - 1, \quad \overline{d-p} = \log\left(\frac{1}{\kappa} - 1\right).$$

Therefore,

$$\kappa_0 \equiv f(\overline{d-p}) - (1-\kappa)\overline{d-p} = -\log \kappa - (1-\kappa) \log\left(\frac{1}{\kappa} - 1\right).$$

We then have $\log(1+e^{d_{t+1}-p_{t+1}}) = f(d_{t+1} - p_{t+1}) \approx \kappa_0 + (1-\kappa)(d_{t+1} - p_{t+1})$, so

$$\begin{aligned} r_{e,t+1} &= p_{t+1} + \log\left(1+e^{d_{t+1}-p_{t+1}}\right) - p_t \\ &\approx p_{t+1} + \kappa_0 + (1-\kappa)(d_{t+1} - p_{t+1}) - p_t \\ &= \kappa_0 + (1-\kappa)d_{t+1} + \kappa p_{t+1} - p_t \\ &= \kappa_0 + \left[(1-\kappa)d_{t+1} - d_t + \kappa p_{t+1} - p_t + d_t\right] \\ &= \kappa_0 + (d_{t+1} - d_t) + \kappa(p_{t+1} - d_{t+1}) - (p_t - d_t) \\ &= \kappa_0 + \Delta d_{t+1} + \kappa(p_{t+1} - d_{t+1}) - (p_t - d_t). \end{aligned}$$

Therefore, we have

$$\begin{aligned} r_{e,t+1} &\approx \kappa_0 + \Delta d_{t+1} + \kappa(p_{t+1} - d_{t+1}) - (p_t - d_t), \\ \kappa &= \frac{1}{1+e^{\overline{d-p}}}, \\ \kappa_0 &= -\log(\kappa) - (1-\kappa) \log\left(\frac{1}{\kappa} - 1\right). \end{aligned}$$

Isolating $p_t - d_t$ and recursively substituting for $p_{t+h} - d_{t+h}$, we have

$$\begin{aligned} (p_t - d_t) &\approx \kappa_0 - r_{e,t+1} + \Delta d_{t+1} + \kappa(p_{t+1} - d_{t+1}) \\ &= \kappa_0 - r_{e,t+1} + \Delta d_{t+1} + \kappa \left[\kappa_0 - r_{e,t+2} + \Delta d_{t+2} + \kappa(p_{t+2} - d_{t+2}) \right] \\ &= (1+\kappa)\kappa_0 + (\Delta d_{t+1} + \kappa \Delta d_{t+2}) - (r_{e,t+1} + \kappa r_{e,t+2}) + \kappa^2(p_{t+2} - d_{t+2}) \\ &\vdots \text{ keep substituting } p_{t+h} - d_{t+h} \\ &= \frac{\kappa_0(1-\kappa^H)}{1-\kappa} + \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} - \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} + \kappa^H(p_{t+H} - d_{t+H}). \end{aligned}$$

Thus, we have

$$d_t - p_t \approx \frac{\kappa_0(\kappa^H - 1)}{1-\kappa} + \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H(d_{t+H} - p_{t+H}),$$

so, letting $dp_t \equiv d_t - p_t$,

$$dp_t \approx \frac{\kappa_0(\kappa^H - 1)}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H dp_{t+H}. \quad (1.1)$$

Taking expectations as of time t ,

$$dp_t = \frac{\kappa_0(\kappa^H - 1)}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] + \kappa^H \mathbb{E}_t[dp_{t+H}]. \quad (1.2)$$

Suppose $H \rightarrow \infty$. Since $\kappa = \frac{1}{1 + e^{\bar{d}-p}} \in (0, 1)$, we have $\lim_{H \rightarrow \infty} \kappa^H = 0$, and by the no-bubble condition $\lim_{H \rightarrow \infty} \kappa^H \mathbb{E}_t[dp_{t+H}] = 0$. Therefore,

$$dp_t = \frac{-\kappa_0}{1 - \kappa} + \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}]. \quad (1.3)$$

1.2 Part (b)



Figure 1: OLS slope coefficients of `future_return`, `change_d`, and `dp_H` on `dp`, plotted against the horizon H (in years).

From the equation

$$1 = \underbrace{\frac{\text{Cov} \left[dp_t, \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} \right]}{\text{Var}[dp]}}_{b_{re}^{(H)}} + \underbrace{\frac{\text{Cov} \left[dp_t, -\sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} \right]}{\text{Var}[dp]}}_{b_{\Delta d}^{(H)}} + \underbrace{\frac{\text{Cov} [dp_t, \kappa^H dp_{t+H}]}{\text{Var}[dp]}}_{b_{dp}^{(H)}}.$$

This equation is trying to measure, how much each component moves when today's dp_t moves. The meaning of these three coefficients is as follows:

$$\sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] = a + b_{r_e}^{(H)} dp_t + \epsilon_t$$

The slope is $b_{r_e}^{(H)}$, and it measures the sensitivity of expected discounted future returns to today's dividend-price ratio.

Because dividend growth enters the Campbell–Shiller equation with a minus sign, define

$$-\sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] = a + b_{\Delta d}^{(H)} dp_t + \epsilon_t.$$

So $b_{\Delta d}^{(H)}$ captures how much variation in dp_t is associated with expected future dividend growth.

$$\kappa^H \mathbb{E}_t[dp_{t+H}] = a + b_{dp}^{(H)} dp_t + \epsilon_t.$$

$b_{dp}^{(H)}$ tells us how much today's high dp_t is simply expected to persist into the future.

$$1 = b_{r_e}^{(H)} + b_{\Delta d}^{(H)} + b_{dp}^{(H)}$$

From the graph, we can see that the contributions of expected future returns, expected future dividend growth, and the future dividend-price ratio to the variation in today's dp_t change with the horizon H . In the short run, the variation in today's dp_t is primarily driven by the future dividend-price ratio. As the horizon H increases, the contribution of expected future returns becomes more significant, while the contribution of the future dividend-price ratio declines toward zero. The contribution of negative expected future dividend growth remains relatively stable across different horizons. Also, if we add up these three coefficients, we get approximately 1, which reflects the normalized variance decomposition of today's dp_t .

1.3 Part (c)



Figure 2: VAR-implied coefficients

The VAR-implied coefficients shown in Figure 2 show some interesting patterns. Instead of the expected future return component alone becoming dominant as H increases, the figure shows that both the expected future return component and the negative expected future dividend growth component contribute significantly to the variation in today's dp_t across different horizons. At the same time, the contribution of the future dividend-price ratio declines as the horizon increases, consistent with the findings from the direct-regression estimates, although the decline appears to be more gradual under the VAR approach. The difference in these patterns may arise because the two approaches estimate long-horizon coefficients differently. The direct approach estimates each horizon separately using realized future outcomes, whereas the VAR approach estimates the joint one-year dynamics of the state variables and uses the estimated transition matrix to imply coefficients at all subsequent horizons.

1.4 Part (d)

The resulting long-run VAR-implied coefficients are reported in Table 1. The table shows that, in the long run, the contributions of expected future returns and negative expected future dividend growth are roughly equal, accounting for approximately 49% and 51%, respectively. Their sum is 0.998808, which is close to, but not exactly, one. This indicates that the unrestricted estimated VAR approximately, rather than mechanically, satisfies the long-run Campbell–Shiller restriction.

Table 1: Long-run VAR-implied coefficients

Component	Coefficient	Estimate
Expected return	$b_{re}^{(\infty)}$	0.489191
Negative expected dividend growth	$b_{dg}^{(\infty)}$	0.509617
Sum	$b_{re}^{(\infty)} + b_{dg}^{(\infty)}$	0.998808

1.5 Part (e)

First, assume $D_t > 0$. We have

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}.$$

Then,

$$\begin{aligned} R_{e,t+1} \frac{D_t}{D_{t+1}} &= \frac{P_{t+1} + D_{t+1}}{P_t} \frac{D_t}{D_{t+1}} \\ &= \frac{D_t}{P_t} \frac{P_{t+1} + D_{t+1}}{D_{t+1}} \\ &= \frac{P_t + D_t}{P_t} \frac{D_t}{P_t + D_t} \\ &= \frac{D_t}{P_{t+1} + D_{t+1}}. \end{aligned}$$

Taking logs on both sides gives

$$\begin{aligned} \log(R_{e,t+1}) + \log\left(\frac{D_t}{D_{t+1}}\right) &= \log\left(\frac{P_t + D_t}{P_t}\right) + \log\left(\frac{D_t}{P_t + D_t}\right) \\ &\quad - \log\left(\frac{D_{t+1}}{P_{t+1} + D_{t+1}}\right). \end{aligned}$$

Define

$$r_{e,t+1} = \log(R_{e,t+1}), \quad p_t = \log(P_t), \quad d_t = \log(D_t), \quad dy_t = \log\left(1 + \frac{D_t}{P_t}\right).$$

Given that

$$\begin{aligned} \log\left(\frac{D_t}{P_t + D_t}\right) &= \log\left(1 - \frac{P_t}{P_t + D_t}\right) \\ &= \log\left(1 - e^{\log(P_t)/(P_t + D_t)}\right) \\ &= \log\left(1 - e^{-\log((P_t + D_t)/P_t)}\right) \\ &= \log\left(1 - e^{-\log(1 + D_t/P_t)}\right) \\ &= \log\left(1 - e^{-dy_t}\right), \end{aligned}$$

we have

$$\log\left(\frac{D_t}{P_t + D_t}\right) = \log\left(1 - e^{-dy_t}\right).$$

Similarly,

$$\log\left(\frac{D_{t+1}}{P_{t+1} + D_{t+1}}\right) = \log\left(1 - e^{-dy_{t+1}}\right).$$

Therefore,

$$r_{e,t+1} - \Delta d_{t+1} = dy_t + \log\left(1 - e^{-dy_t}\right) - \log\left(1 - e^{-dy_{t+1}}\right).$$

Let

$$f(x) = \log(1 - e^{-x}).$$

Then,

$$f'(x) = \frac{1}{1 - e^{-x}} e^{-x} = \frac{e^{-x}}{1 - e^{-x}}.$$

Let $x = dy_t$. By a first-order approximation, and defining $\kappa \equiv e^{-\bar{dy}}$, we have

$$\begin{aligned} \log(1 - e^{-dy_t}) &\approx \log(1 - e^{-\bar{dy}}) + \log'(1 - e^{-\bar{dy}})(dy_t - \bar{dy}) \\ &= \log(1 - \kappa) + \frac{e^{-\bar{dy}}}{1 - e^{-\bar{dy}}}(dy_t - \bar{dy}) \\ &= \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_t - \bar{dy}). \end{aligned}$$

Thus,

$$\log(1 - e^{-dy_t}) \approx \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_t - \bar{dy}).$$

Similarly,

$$\log(1 - e^{-dy_{t+1}}) \approx \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_{t+1} - \bar{dy}).$$

Substituting these approximations into $r_{e,t+1} - \Delta d_{t+1} = dy_t + \log(1 - e^{-dy_t}) - \log(1 - e^{-dy_{t+1}})$, we obtain

$$\begin{aligned} r_{e,t+1} - \Delta d_{t+1} &= dy_t + \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_t - \bar{dy}) \\ &\quad - \log(1 - \kappa) - \frac{\kappa}{1 - \kappa}(dy_{t+1} - \bar{dy}) \\ &= dy_t + \frac{\kappa}{1 - \kappa}(dy_t - dy_{t+1}) \\ &= \frac{1}{1 - \kappa}dy_t - \frac{\kappa}{1 - \kappa}dy_{t+1}. \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{1}{1 - \kappa}dy_t &= r_{e,t+1} - \Delta d_{t+1} + \frac{\kappa}{1 - \kappa}dy_{t+1}, \\ dy_t &= (1 - \kappa)(r_{e,t+1} - \Delta d_{t+1}) + \kappa dy_{t+1}. \end{aligned}$$

Recursively substituting for future values of dy , we have

$$\begin{aligned} dy_t &= (1 - \kappa)(r_{e,t+1} - \Delta d_{t+1}) \\ &\quad + \kappa[(1 - \kappa)(r_{e,t+2} - \Delta d_{t+2}) + \kappa dy_{t+2}] \\ &\quad \vdots \quad \text{keep substituting } dy_{t+h} \\ &= (1 - \kappa) \left(\sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} \right) + \kappa^H dy_{t+H}. \end{aligned}$$

Thus, we have

$$dy_t = (1 - \kappa) \left(\sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} \right) + \kappa^H dy_{t+H}.$$

Taking expectations at time t , we have

$$dy_t = (1 - \kappa) \left(\sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] \right) + \kappa^H \mathbb{E}_t[dy_{t+H}].$$

Suppose $H \rightarrow \infty$. Given that $\kappa = e^{-\bar{d}y} \in (0, 1)$, we have

$$\lim_{H \rightarrow \infty} \kappa^H = 0.$$

Also, because of the no-bubble condition,

$$\lim_{H \rightarrow \infty} \kappa^H \mathbb{E}_t[dy_{t+H}] = 0.$$

Therefore,

$$dy_t = (1 - \kappa) \left(\sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] \right).$$

2 Question 2

2.1 Part (a)

Figure 3 shows that the adjusted R^2 generally rises with the horizon, from about 4.6% at one year to a peak of about 44.3% at 14 years, before declining slightly to about 42.1% at 15 years. Thus, the current dividend-price ratio has substantially greater explanatory power for long-horizon average excess returns than for one-year-ahead excess returns.



Figure 3: Adjusted R^2 from predictive regressions of average future annual simple excess returns on the starting dividend-price ratio.

2.2 Part (b)

I estimate the following predictive regression by OLS:

$$xR_{e,t+1} = a + b \frac{D_t}{P_t} + \varepsilon_t,$$

where $xR_{e,t+1}$ is the annual excess equity return and D_t/P_t is the current dividend-price ratio.

To examine the sensitivity of statistical inference to the choice of covariance estimator, I calculate five sets of standard errors. In addition to conventional OLS and heteroskedasticity-robust White standard errors, I use Newey–West and Hansen–Hodrick estimators with 11 lags to account for autocorrelation induced by overlapping annual returns. For the automatic Newey–West estimator, I use the Newey and West (1994) plug-in bandwidth selection procedure, which yields a bandwidth of 22.0776 and a lag truncation of $L = \lfloor 22.0776 \rfloor = 22$.

In Table 2, the estimated slope is positive, suggesting that the dividend-price ratio predicts higher future excess equity returns. While the coefficient is highly significant under conventional OLS and White standard errors, its statistical significance becomes weaker when accounting for serial correlation for the Hansen–Hodrick and Newey–West estimates.

Table 2: Q2(b) slope estimate and alternative standard errors

Method	$\hat{b}$	SE($\hat{b}$)	t -statistic	Lags/bandwidth
Baseline OLS	2.8038	0.3802	7.3754	0
White (HC0)	2.8038	0.6587	4.2569	0
Newey–West (1987)	2.8038	1.2990	2.1584	11
Hansen–Hodrick (1980)	2.8038	1.4537	1.9287	11
Newey–West (1994), automatic	2.8038	1.2523	2.2390	22.0776 ($L = 22$)

2.3 Part (c)

From Table 3, the Amihud and Hurvich estimate of b is 2.3412, which is smaller than the standard OLS estimate of 2.8038. It is natural for the two estimates to differ because as indicated by the estimated persistence coefficient $\hat{\phi} = 0.7197$, the predictor D_t/P_t is persistent. In a finite sample, persistence in the predictor can induce small-sample bias in the OLS estimate of the predictive-regression coefficient. The Amihud and Hurvich procedure corrects for this problem by first bias-correcting the persistence coefficient from $\hat{\phi} = 0.7197$ to $\hat{\phi}^c = 0.7540$ and then including the resulting bias-corrected residual $\hat{u}_{t+1}^c$ in the predictive regression. Thus, the difference between $\hat{b}_{\text{AH}}$ and $\hat{b}_{\text{OLS}}$ is expected.

Table 3: Q2(c) Amihud–Hurvich estimates

Quantity	Estimate
$\hat{\theta}$	0.0111
$\hat{\phi}$	0.7197
T (years)	95
$\hat{\phi}^c$	0.7540
$\hat{b}_{\text{AH}}$	2.3412
$\hat{b}_u$	−13.4837
$\hat{b}_{\text{OLS}}$	2.8038
$\hat{b}_{\text{AH}} - \hat{b}_{\text{OLS}}$	−0.4626
Observations	1,117

2.4 Part (d)

I estimate Equation (2.2) out of sample using an expanding window. The first out-of-sample forecast corresponds to December 1940, using the dividend-price ratio observed in December 1939. At each forecast date, I re-estimate the predictive regression using the historical information available at that date and use the resulting coefficients to construct the out-of-sample forecast. I also calculate the historical expanding-window mean excess return, $\overline{xR}_{e,t}$, which serves as the benchmark forecast. The in-sample fitted values are obtained from the full-sample regression in Equation (2.2).

The out-of-sample R^2 is calculated as

$$R_{OS}^2 = 1 - \frac{\sum_t \left(xR_{e,t+1} - \hat{E}_t^{OS}[xR_e] \right)^2}{\sum_t \left(xR_{e,t+1} - \overline{xR}_{e,t} \right)^2}.$$

The R_{OS}^2 compares the forecast errors from the predictive regression with those from the historical expanding-window mean benchmark. A positive R_{OS}^2 indicates that the dividend-price-ratio forecast has a lower sum of squared forecast errors than the historical-mean forecast, whereas a negative value indicates that the historical mean performs better.

For the full out-of-sample period, I obtain $R_{OS}^2 = -0.0061$. Thus, over the full evaluation period, the dividend-price-ratio forecast performs slightly worse than the historical-mean benchmark.

To examine variation in forecasting performance over time, I also calculate R_{OS}^2 using a 50-year (600-month) rolling window of forecast errors. The coefficients a_τ and b_τ continue to be estimated using the expanding window. Thus, the 50-year rolling window changes only the forecast errors included in the calculation of R_{OS}^2 ; it does not change how the forecasts themselves are constructed.

Figure 5 shows substantial variation in out-of-sample forecasting performance over time. The rolling R_{OS}^2 is strongly positive in the early part of the sample, indicating that the dividend-price-ratio forecast outperforms the historical-mean benchmark over those 50-year windows. It subsequently declines and fluctuates around zero for much of the later sample, before becoming increasingly negative toward the end.

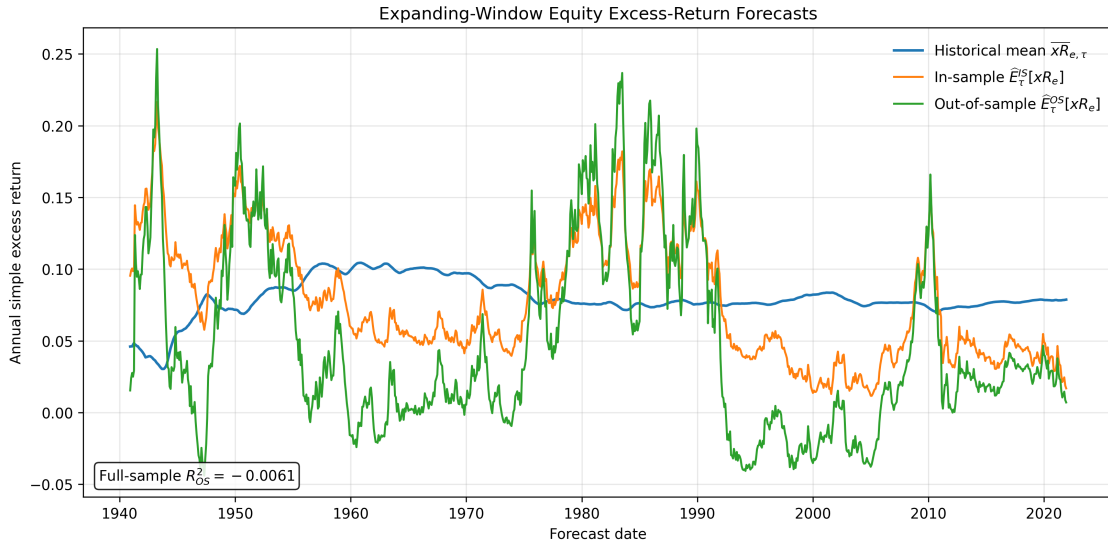


Figure 4: Historical-mean, in-sample, and expanding-window out-of-sample forecasts of the annual simple equity excess return.



Figure 5: Fifty-year rolling out-of-sample R^2 .

2.5 Part (e)

I repeat the out-of-sample exercise in Part (d), but impose the steady-state restrictions on the forecasting coefficients. For each forecast date τ , I calculate

$$G_\tau = \text{mean} \left(e^{dgs} \right)$$

using the same expanding historical sample used to calculate the historical-mean forecast $\overline{xR}_{e,\tau}$. I then impose

$$\hat{a}_\tau = G_\tau - 1, \quad \hat{b}_\tau = G_\tau,$$

so that the restricted out-of-sample forecast is

$$\hat{E}_\tau^{OS}[xR_e] = (G_\tau - 1) + G_\tau DP_\tau.$$

Thus, unlike Part (d), the coefficients are not estimated by OLS at each forecast date; instead, they are determined by the steady-state valuation restrictions.

Figure 6 compares the restricted out-of-sample forecast with the historical-mean forecast and the full-sample in-sample fitted values. Over the full out-of-sample evaluation period, the restricted forecast produces

$$R_{OS}^2 = 0.0321.$$

Since this value is positive, the restricted steady-state forecast has a lower sum of squared forecast errors than the historical-mean benchmark over the full out-of-sample period.

Figure 7 reports the 50-year rolling R_{OS}^2 . As in Part (d), the rolling window is applied only to the forecast errors used to calculate R_{OS}^2 ; G_τ continues to be calculated using the expanding historical window. The rolling R_{OS}^2 remains positive throughout the plotted period, indicating that the restricted forecast outperforms the historical-mean benchmark over each of these 50-year evaluation windows. However, its magnitude varies over time and declines substantially toward the end of the sample, indicating that the degree of out-of-sample improvement is not constant over time.



Figure 6: Historical-mean, in-sample, and restricted steady-state out-of-sample forecasts of the annual simple equity excess return.



Figure 7: Fifty-year rolling out-of-sample R^2 for the restricted forecast.

3 Question 3

3.1 Part (a)

I construct the monthly momentum signal for each stock using CRSP returns. Momentum at the end of month τ is the cumulative return over the preceding 12 months, excluding the current month:

$$MOM_{i,\tau} = \prod_{k=1}^{12} (1 + RET_{i,\tau-k}) - 1$$

I require 12 valid monthly returns in consecutive calendar months. The sample begins in June 1963 and is restricted to common stocks with share codes 10 or 11 listed on NYSE, Amex or Nasdaq. I exclude utilities and financial firms.

To validate the constructed momentum signal, I merge the CRSP observations with the Chen-Zimmermann dataset. I then estimate the following cross-sectional regression separately for each month:

$$MOMCZ_{i,\tau} = a_{\tau} + b_{\tau}MOM_{i,\tau} + \varepsilon_{i,\tau}.$$

Figures 8–10 plot the monthly intercept estimates, slope estimates and (R^2) values, respectively.

The estimated intercepts generally fluctuate around zero. The slopes generally remain close to one. The (R^2) values are consistently high and close to one. The results suggest that the CRSP-based momentum signal closely matches the Chen–Zimmermann measure.

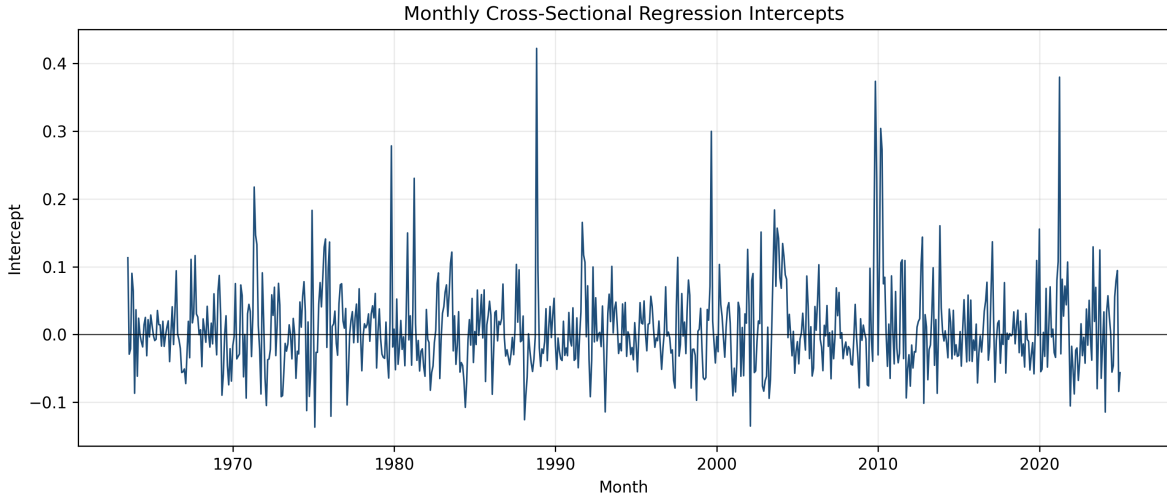


Figure 8: Monthly cross-sectional regression intercepts.



Figure 9: Monthly cross-sectional regression slopes.



Figure 10: Monthly cross-sectional regression R^2 values.

3.2 Part (b)

I construct the book-to-market ratio for each stock using CRSP and Compustat data. For the portfolio formed at the end of June of year t , I calculate

$$BM_{i,t} = \frac{BE_{i,t-1}}{ME_{i,Dec(t-1)}},$$

where market equity is calculated as $ME = |PRC| \times SHROUT$, and book equity is calculated as $BE = SE + TXDITC - BVPS$, using the specified Compustat variable hierarchy. The resulting BM signal is held constant from June of year t through May of year $t + 1$.

I require at least two previous years of Compustat records and positive book equity. I restrict the sample to common stocks listed on NYSE, Amex, or Nasdaq, excluding utilities and financial firms. The sample begins in June 1963.

To compare my constructed BM signal with the Chen–Zimmermann measure, I merge the two datasets by stock and month and estimate the following cross-sectional regression separately for each month:

$$BM_{CZ,i,\tau} = a_\tau + b_\tau BM_{i,\tau} + \varepsilon_{i,\tau},$$

Figures 11–13 report the monthly intercepts, slopes, and R^2 values, respectively. The results indicate that the constructed BM signal closely matches the Chen–Zimmermann measure. Across the sample period except for June 2024, the intercepts are generally close to zero, the slopes are generally close to one, and the regressions exhibit high explanatory power.

As a robustness check, I investigate the sharp decline in the regression statistics observed in June 2024. Further examination identifies a highly influential observation, PERMNO 18558, which has exceptionally large book-to-market values and therefore substantial leverage in the cross-sectional regression. To assess whether the results are driven by this extreme observation, I winsorize both my constructed BM measure and the Chen–Zimmermann BM measure separately at the 1st and 99th percentiles within each month and re-estimate the monthly regressions. After winsorization, from the Figure 14–16, the results suggest that the two BM measures remain closely aligned after reducing the influence of extreme observations.

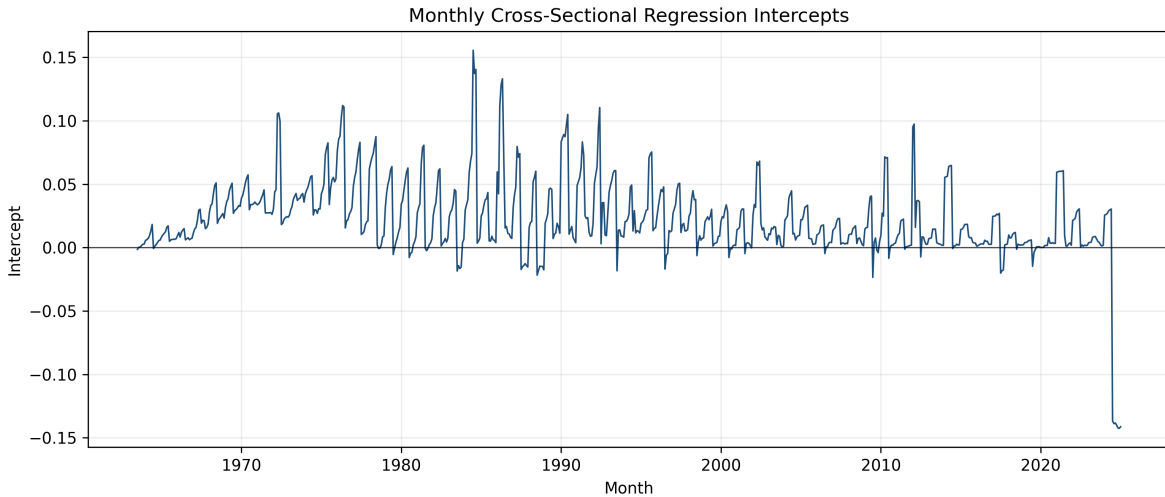


Figure 11: Monthly cross-sectional regression intercepts.

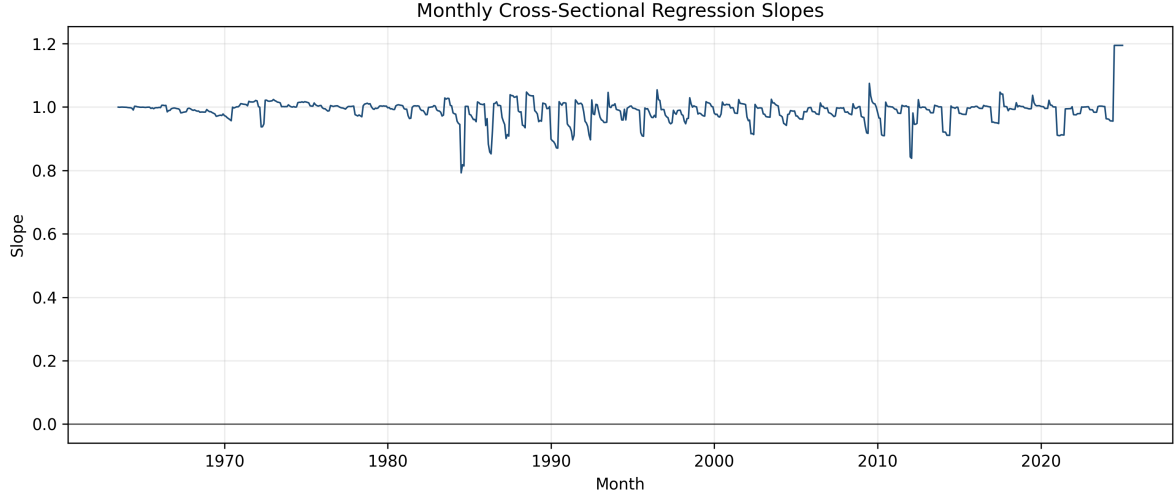


Figure 12: Monthly cross-sectional regression slopes.

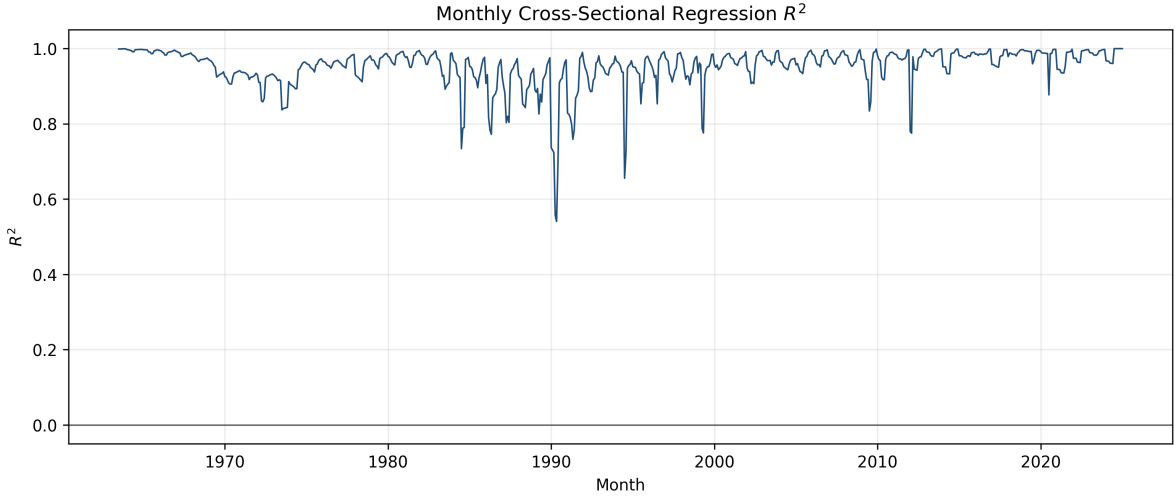


Figure 13: Monthly cross-sectional regression R^2 values.

1% winsorized robustness check. The full-sample June 2024 regression is dominated by PERMNO 18558, whose exceptionally large book-to-market values give it very high cross-sectional leverage. As a sensitivity analysis, I winsorize both constructed BM and BM_{CZ} separately within each month at their 1st and 99th percentiles before estimating the same monthly regressions. These robustness results do not replace the full-sample estimates above.

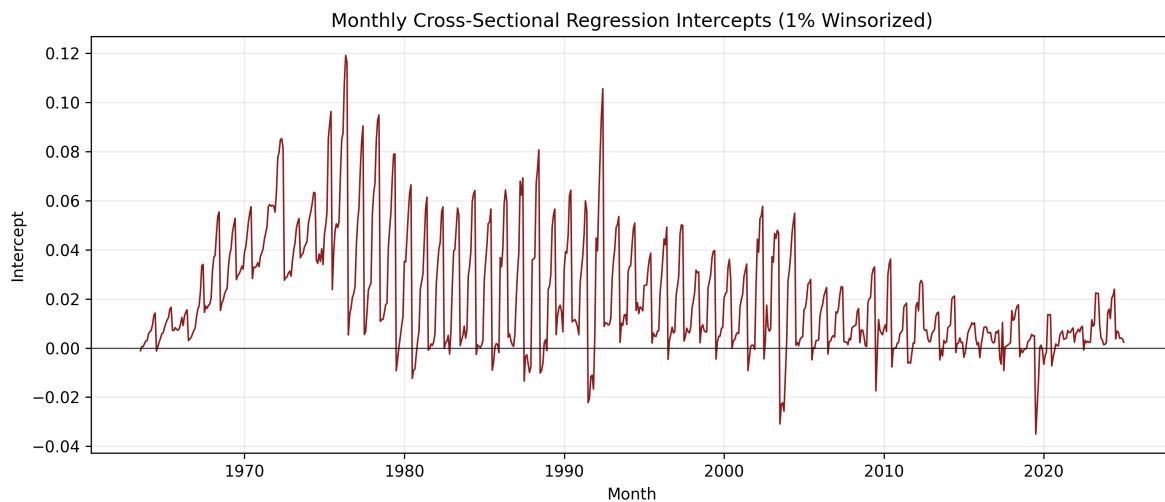


Figure 14: Monthly cross-sectional regression intercepts after 1% winsorization of both BM measures.

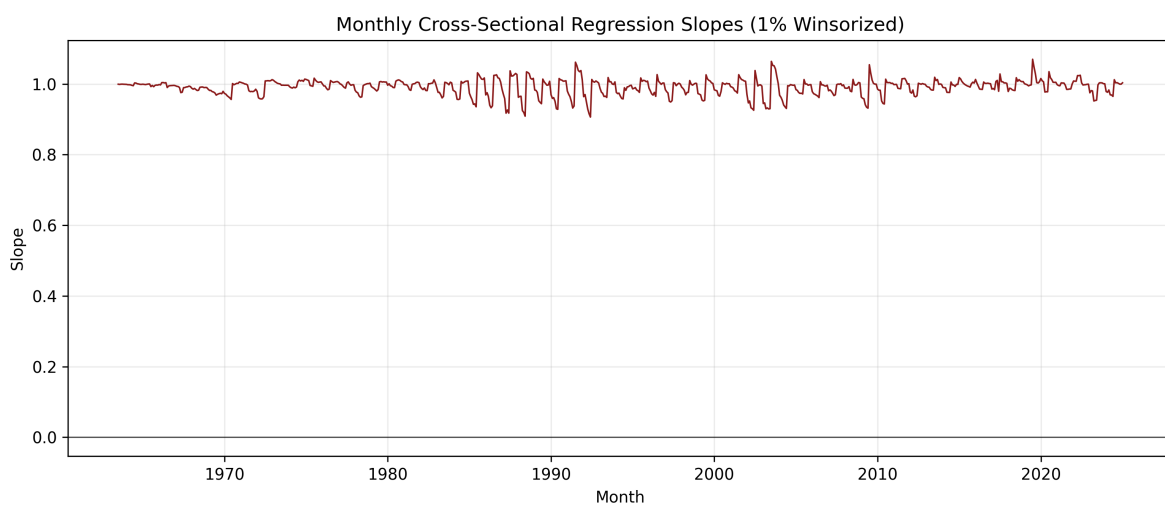


Figure 15: Monthly cross-sectional regression slopes after 1% winsorization of both BM measures.

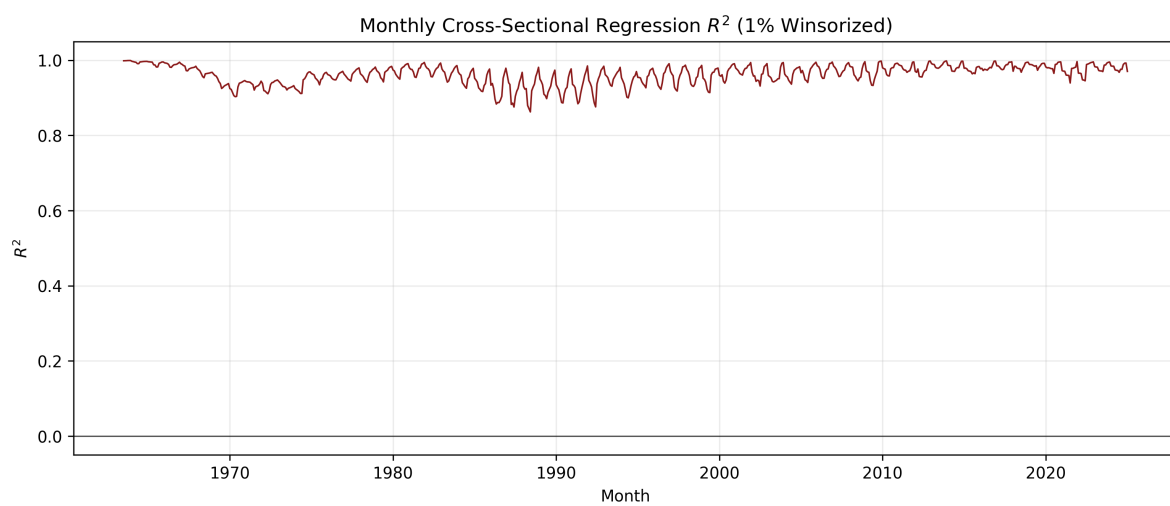


Figure 16: Monthly cross-sectional regression R^2 values after 1% winsorization of both BM measures.

3.3 Part (c)

Figures 17–21 present the average monthly excess returns of decile portfolios formed using the three Chen–Zimmermann signals: book-to-market (BM), momentum (MOM), and gross profitability (GP). The five portfolio-construction methods differ in their weighting schemes, rebalancing frequencies, and breakpoint definitions. Table 4 reports the corresponding high-minus-low (HML) average monthly returns and Newey–West t-statistics.

Several patterns emerge from the figures. First, BM generally exhibits a positive relationship between signal rankings and average excess returns. This pattern is particularly pronounced in the equal-weighted portfolios (Figures 18 and 21), whereas the differences between low- and high-BM deciles are smaller in the value-weighted portfolios (Figures 17, 19, and 20).

Second, momentum exhibits greater sensitivity to portfolio-construction methods. The return spread between high- and low-momentum deciles is particularly pronounced under monthly value-weighted rebalancing with NYSE breakpoints (Figure 19). In contrast, the relationship is weaker under annual rebalancing. In the equal-weighted, annually rebalanced portfolios with NYSE breakpoints (Figure 18), the highest-momentum decile has a slightly lower average excess return than the lowest-momentum decile.

Third, GP generally exhibits higher average excess returns in the high-signal deciles than in the low-signal deciles across all five portfolio-construction methods. However, the differences are relatively modest in the value-weighted portfolios and more pronounced in the equal-weighted portfolios.

Comparing the five figures, portfolio weighting and rebalancing frequency appear to have substantial effects on the observed return patterns. These patterns are also reflected in Table 4. BM and GP have positive HML returns across all five methods, whereas the magnitude and sign of the momentum HML return vary with portfolio construction.

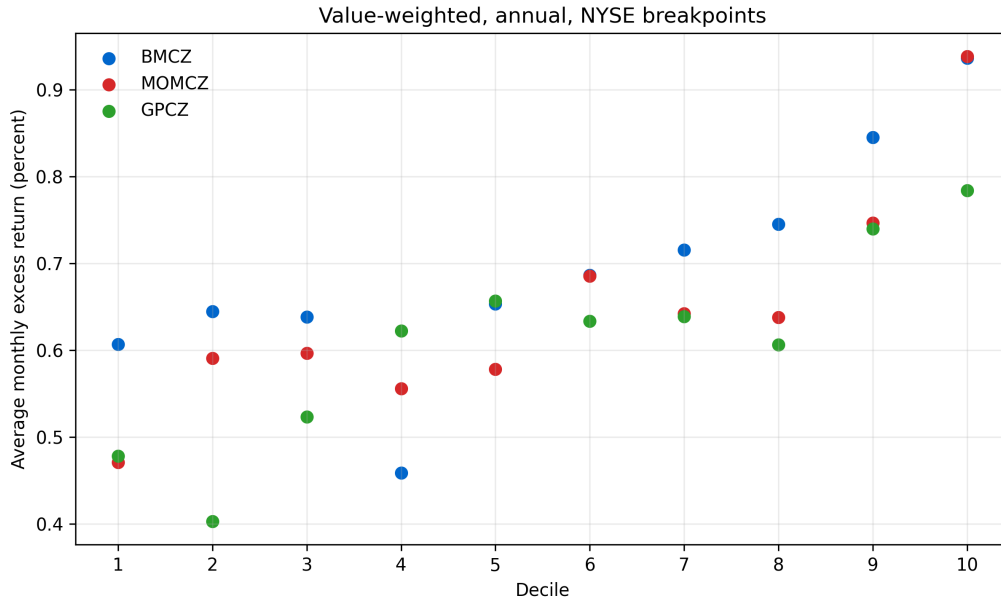


Figure 17: Average monthly excess returns: value-weighted annual portfolios with NYSE breakpoints.

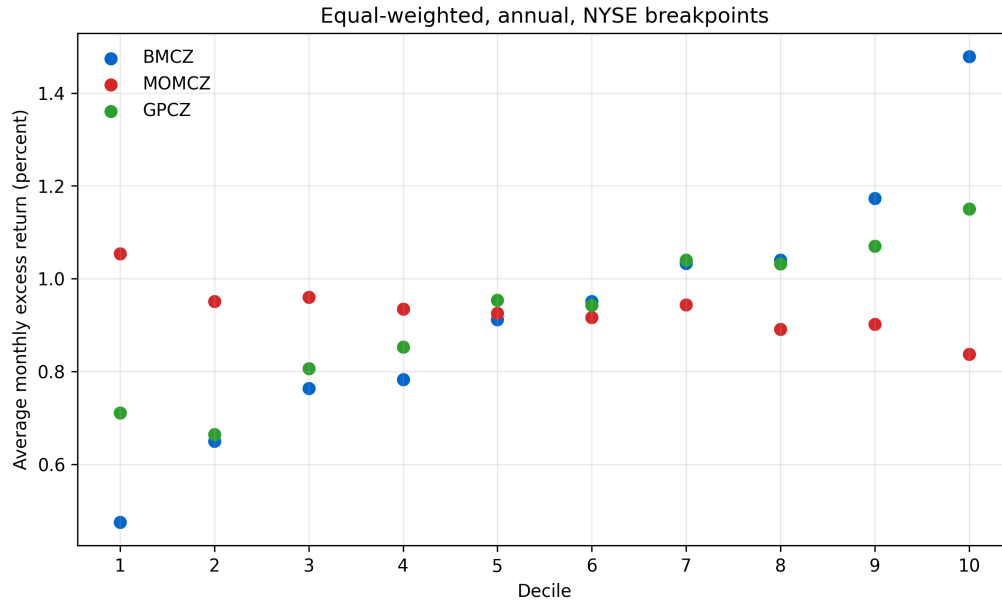


Figure 18: Average monthly excess returns: equal-weighted annual portfolios with NYSE breakpoints.

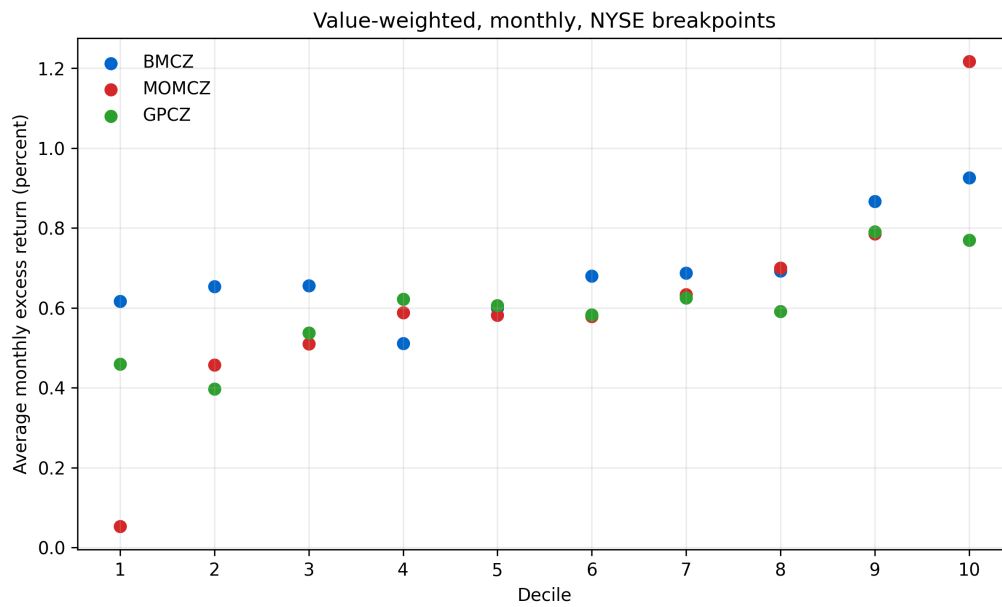


Figure 19: Average monthly excess returns: value-weighted monthly portfolios with NYSE breakpoints.

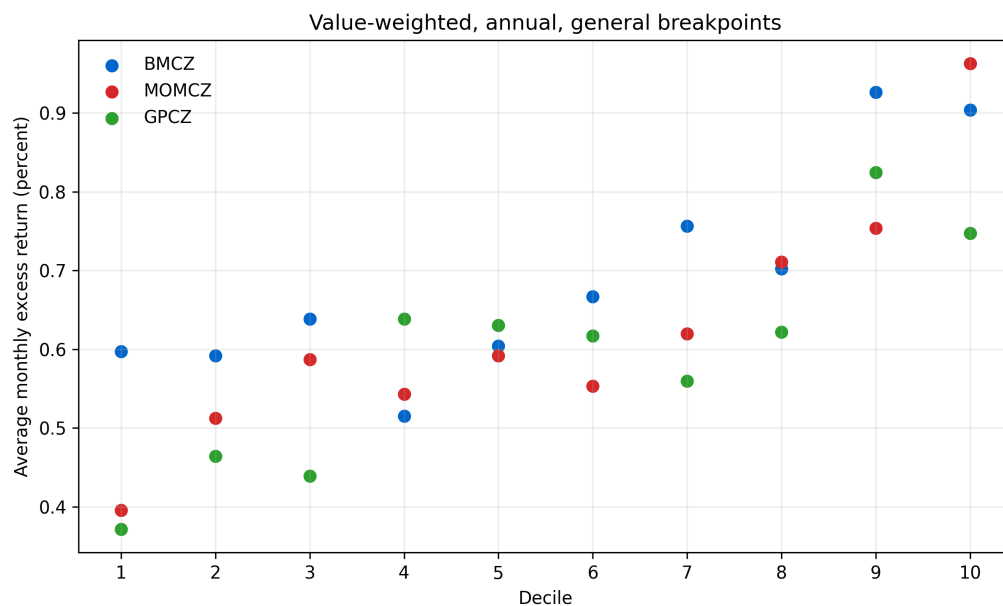


Figure 20: Average monthly excess returns: value-weighted annual portfolios with general breakpoints.

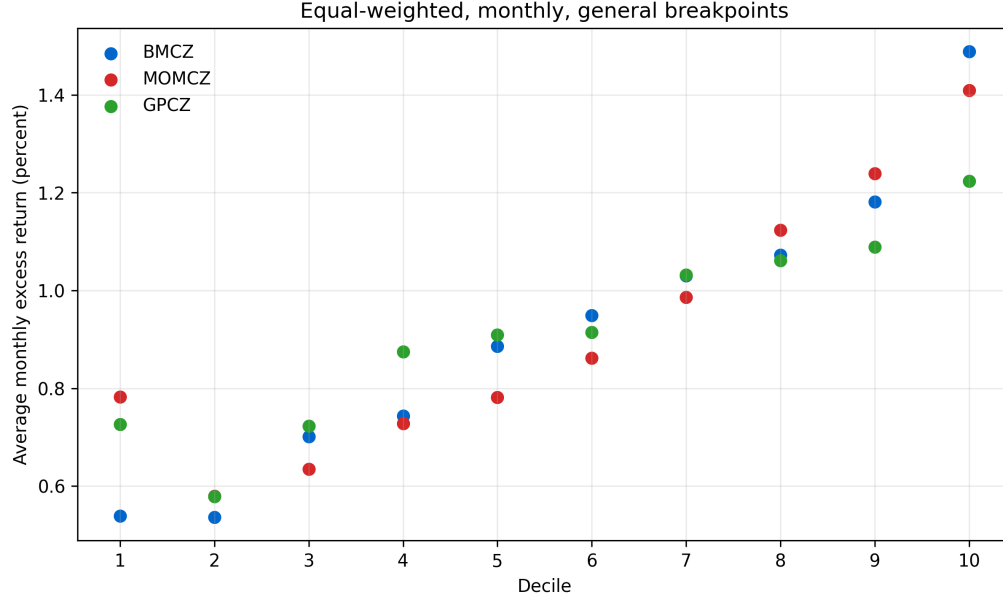


Figure 21: Average monthly excess returns: equal-weighted monthly portfolios with general breakpoints.

Table 4: Average monthly HML returns and Newey–West t -statistics

Portfolio construction	Signal	Mean (%)	NW t -statistic
VW annual, NYSE	BMCZ	0.3295	1.579
VW annual, NYSE	MOMCZ	0.4672	1.979
VW annual, NYSE	GPCZ	0.3059	1.932
EW annual, NYSE	BMCZ	1.0030	5.751
EW annual, NYSE	MOMCZ	−0.2164	−1.063
EW annual, NYSE	GPCZ	0.4393	2.856
VW monthly, NYSE	BMCZ	0.3087	1.494
VW monthly, NYSE	MOMCZ	1.1644	4.307
VW monthly, NYSE	GPCZ	0.3114	1.969
VW annual, general	BMCZ	0.3065	1.352
VW annual, general	MOMCZ	0.5674	2.035
VW annual, general	GPCZ	0.3758	1.873
EW monthly, general	BMCZ	0.9496	5.266
EW monthly, general	MOMCZ	0.6273	2.228
EW monthly, general	GPCZ	0.4976	2.545

3.4 Part (d)

In Table 5, the Fama–MacBeth regressions reveal positive relationships between future stock excess returns and both book-to-market and gross profitability, while equity duration is negatively associated with future excess returns.

Under OLS, all three characteristics are statistically significant at the 1% level in their respective univariate regressions. When all three characteristics are included simultaneously, their coefficients remain statistically significant at the 1% level, suggesting that each contains incremental information about the cross-section of future stock returns.

The WLS results are different. The book-to-market coefficient becomes substantially smaller and is generally statistically insignificant. Similarly, the negative duration coefficient loses statistical significance in the full specification. In contrast, gross profitability remains positively associated with future excess returns and is significant at the 5% level in the full WLS specification.

Overall, the results indicate that the estimated relationships between stock characteristics and future returns are sensitive to market-equity weighting.

Table 5: Fama–MacBeth regressions of next-month stock excess returns: conventional inference

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Panel A: OLS</i>							
Intercept	0.5278** (2.04)	0.7207*** (2.58)	1.6800*** (7.28)	0.0138 (0.05)	1.4001*** (6.09)	1.4832*** (5.67)	0.7432*** (2.80)
Q^{BM}	0.0929*** (4.90)			0.1079*** (5.63)	0.0333* (1.67)		0.0619*** (2.77)
Q^{GP}		0.0576*** (4.40)		0.0785*** (6.03)		0.0252* (1.87)	0.0525*** (3.55)
Q^{Dur}			−0.1168*** (−6.65)		−0.0990*** (−5.19)	−0.1062*** (−5.67)	−0.0607*** (−2.75)
Months	618	618	618	618	618	618	618
<i>Panel B: WLS</i>							
Intercept	0.6411*** (3.06)	0.4879** (2.13)	1.3349*** (6.35)	0.2187 (0.83)	1.4015*** (4.85)	0.8396** (2.54)	0.6538* (1.68)
Q^{BM}	0.0188 (0.77)			0.0511** (2.09)	−0.0255 (−0.91)		0.0146 (0.59)
Q^{GP}		0.0337* (1.78)		0.0522*** (2.99)		0.0474** (2.02)	0.0519** (2.23)
Q^{Dur}			−0.0842*** (−3.53)		−0.0848*** (−2.96)	−0.0505* (−1.67)	−0.0365 (−1.06)
Months	618	618	618	618	618	618	618

Notes: Coefficients are monthly percentage points. Q^{BM} , Q^{GP} , and Q^{Dur} are monthly cross-sectional deciles from 1 to 10. WLS uses month- τ market equity as the weight. Conventional Fama–MacBeth t -statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Newey–West-adjusted results are reported separately as a robustness check in my Github repository output/Q3d_FM_NeweyWest.tex.

3.5 Part (e)

Table 6 reports pooled OLS regressions of monthly portfolio excess returns on portfolio-level BM, GP, and duration decile ranks. Following the assignment, I estimate the seven specifications separately for value-weighted and equal-weighted portfolios. The portfolios are rebalanced annually using June NYSE breakpoints. For each portfolio-month, I calculate the average decile rank of its constituent stocks, using the same weights as for the portfolio return. Specifically, for value-weighted portfolios, I calculate each portfolio-level decile rank as the market-equity-weighted average of its constituent stocks' decile ranks; for equal-weighted portfolios, I use the arithmetic average. Driscoll–Kraay standard errors are used for inference.

In the univariate regressions, BM is positive and statistically significant for equal-weighted portfolios but not for value-weighted portfolios. GP is positive and significant at the 1% level for equal-weighted portfolios, but is not significant for value-weighted portfolios. Duration has a negative and statistically significant coefficient in both panels. In the joint BM–GP regressions, both coefficients are positive and significant in both panels. When duration is included, the BM and GP coefficients are generally insignificant. In the full specification, only duration is statistically significant for value-weighted portfolios; none of the three characteristics is significant for equal-weighted portfolios. The negative univariate relationship between duration and subsequent excess returns is consistent with the Fama–MacBeth results in Question 3(d).

The full-specification results differ: all three characteristics are significant in the stock-level OLS regression, only GP is significant in the stock-level WLS regression, and only duration is significant in the value-weighted portfolio regression. None is significant in the equal-weighted portfolio regression.

Table 6: Portfolio-panel regressions with Driscoll–Kraay inference

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Panel A: Value-weighted portfolios</i>							
Intercept	0.5682*** (2.72)	0.5359** (2.27)	1.2322*** (5.98)	−0.2405 (−0.52)	1.3082*** (4.23)	1.4096*** (4.14)	1.5097* (1.86)
Dec^{BM}	0.0393 (1.45)			0.1048** (2.23)	0.0005 (0.01)		−0.0081 (−0.13)
Dec^{GP}		0.0272 (1.15)		0.0970** (2.21)		−0.0048 (−0.18)	−0.0050 (−0.10)
Dec^{Dur}			−0.0704*** (−3.03)		−0.0869*** (−2.76)	−0.1019*** (−3.01)	−0.1122** (−2.08)
Observations	6180	6180	6180	12360	12360	12360	18540
<i>Panel B: Equal-weighted portfolios</i>							
Intercept	0.5414** (2.06)	0.7765*** (2.61)	1.5448*** (6.24)	0.0196 (0.06)	1.0221*** (3.20)	1.5733*** (4.95)	0.7423 (1.08)
Dec^{BM}	0.0905*** (3.48)			0.1090*** (4.08)	0.0635 (1.60)		0.0712 (1.30)
Dec^{GP}		0.0477*** (2.79)		0.0761*** (4.32)		−0.0010 (−0.05)	0.0354 (0.92)
Dec^{Dur}			−0.0922*** (−4.74)		−0.0600* (−1.83)	−0.0967*** (−4.25)	−0.0527 (−1.03)
Observations	6180	6180	6180	12360	12360	12360	18540

Notes: Coefficients are monthly percentage points. The panel contains 30 annually rebalanced portfolios formed at June using NYSE breakpoints: ten each for BM, GP, and duration. Stock-level monthly cross-sectional decile assignments are identical in both panels. In Panel A, portfolio returns and Dec^{BM} , Dec^{GP} , and Dec^{Dur} use month- τ market-equity weights. In Panel B, portfolio returns and all three portfolio-level decile measures use equal constituent weights (arithmetic averages). Pooled OLS is estimated separately by panel. Driscoll–Kraay t -statistics using a Bartlett kernel and automatic bandwidth are reported in parentheses. ***, **, and * denote two-sided significance at the 1%, 5%, and 10% levels, respectively.

4 Question 4

4.1 Part (a)

I use the Fama–Bliss yields for maturities from one to five years. Since the original yields are reported in percentage terms, I first divide them by 100 and then construct the log yields, log forward rates, and log annual returns according to the formulas in the question. Because the data are monthly, I use a 12-month lag when constructing annual returns.

For each maturity $H = 2, 3, 4, 5$, I calculate the excess log yield, excess log forward rate, and excess log annual return relative to the corresponding one-year value. Table 7 reports their time-series averages in percentage terms, obtained by multiplying the averages in log units by 100.

Table 7: Time-series averages of bond excess log yields, forward rates, and annual returns (percent)

H	$\overline{xy}_b^{(H)}$	$\overline{xf}_b^{(H)}$	$\overline{xr}_b^{(H)}$
2	0.168613	0.337226	0.315423
3	0.327787	0.646136	0.606929
4	0.465958	0.880470	0.819825
5	0.563911	0.955723	0.871854

4.2 Part (b)

For $H = 2, 3, 4, 5$, I estimate the following regression by OLS:

$$\frac{1}{H}xr_{b,t \rightarrow t+H}^{(H)} = a^{(H)} + b^{(H)}xy_{b,t}^{(H)} + \epsilon_t^{(H)},$$

where $xy_{b,t}^{(H)} = y_{b,t}^{(H)} - y_{b,t}^{(1)}$. The dependent variable is the annualized hold-to-maturity excess log return relative to rolling over one-year bonds. Hansen–Hodrick standard errors account for overlapping holding periods.

I used a common sample period of 811 monthly observations for all four maturities, from June 1952 through December 2019.

The estimated coefficients are positive and statistically significant at the 5% level for all four maturities. This indicates that higher current yield spreads predict higher future annualized excess bond returns. The coefficients decline from 0.6931 for two-year bonds to 0.3140 for five-year bonds, while R^2 decreases from 8.09% to 2.81%. Thus, yield spreads have modest predictive power, which declines with maturity in this sample. These findings provide evidence against the simple expectations hypothesis and are consistent with time-varying bond risk premia.

Table 8: Hold-to-maturity excess bond return regressions

	$H = 2$	$H = 3$	$H = 4$	$H = 5$
$\widehat{b}^{(H)}$ [HH t -stat]	0.6931 [3.4373]	0.5201 [2.9990]	0.3935 [2.2761]	0.3140 [2.0716]
R^2	8.0851%	5.4333%	3.7891%	2.8126%
N	811	811	811	811

4.3 Part (c)

For $H = 2, 3, 4, 5$, I estimate the following regression by OLS:

$$xr_{b,t+1}^{(H)} = a^{(H)} + b^{(H)} xf_{b,t}^{(H)} + \epsilon_t^{(H)},$$

where $xf_{b,t}^{(H)} = f_{b,t}^{(H)} - f_{b,t}^{(1)}$ is the current excess log forward rate. I calculate t-statistics using Newey–West standard errors with automatic lag selection.

I used the same common sample as in Question 4(b): 811 monthly observations for all four maturities, from June 1952 through December 2019. Consistent with Footnote 15, the estimated $H = 2$ slope and R^2 are therefore identical in Questions 4(b) and 4(c).

In Table 9, all four estimated coefficients are positive and statistically significant at the 5% level, indicating that higher current forward spreads predict higher one-year-ahead excess bond returns. The R^2 values range from 6.85% to 11.48%, with the highest explanatory power for four-year bonds. These findings provide evidence against the simple expectations hypothesis and are consistent with time-varying expected bond risk premia.

Table 9: One-year excess bond return regressions on forward spreads

	2 years	3 years	4 years	5 years
<i>Forwards</i>				
$\widehat{b}^{(H)}$ [NW t -stat]	0.6931 [3.0524]	0.9028 [3.0669]	1.1476 [3.4600]	0.9715 [2.7420]
R^2	8.0851%	8.8916%	11.4812%	6.8536%
N	811	811	811	811

4.4 Part (d)

I estimate the Cochrane–Piazzesi factor by regressing the average one-year-ahead excess return across the two- to five-year bonds on the current one- to five-year forward rates:

$$\frac{1}{4} \sum_{H=2}^5 x r_{b,t+1}^{(H)} = \theta_0 + \sum_{h=1}^5 \theta_h f_{b,t}^{(h)} + u_t.$$

Using the estimated slope coefficients, I construct

$$cp_t = \sum_{h=1}^5 \hat{\theta}_h f_{b,t}^{(h)}.$$

Figure 22 plots the estimated factor in percentage terms, with NBER recessions shown by the shaded areas. The CP factor varies considerably over time, suggesting substantial time variation in expected excess bond returns. Although the factor sometimes moves sharply around recessions, it does not follow the same pattern in every recession. The factor reaches its highest levels during the first half of the 1980s and remains relatively elevated in the early 1990s and early 2000s. It is comparatively low around 2020 and becomes negative during parts of 2022–2024.

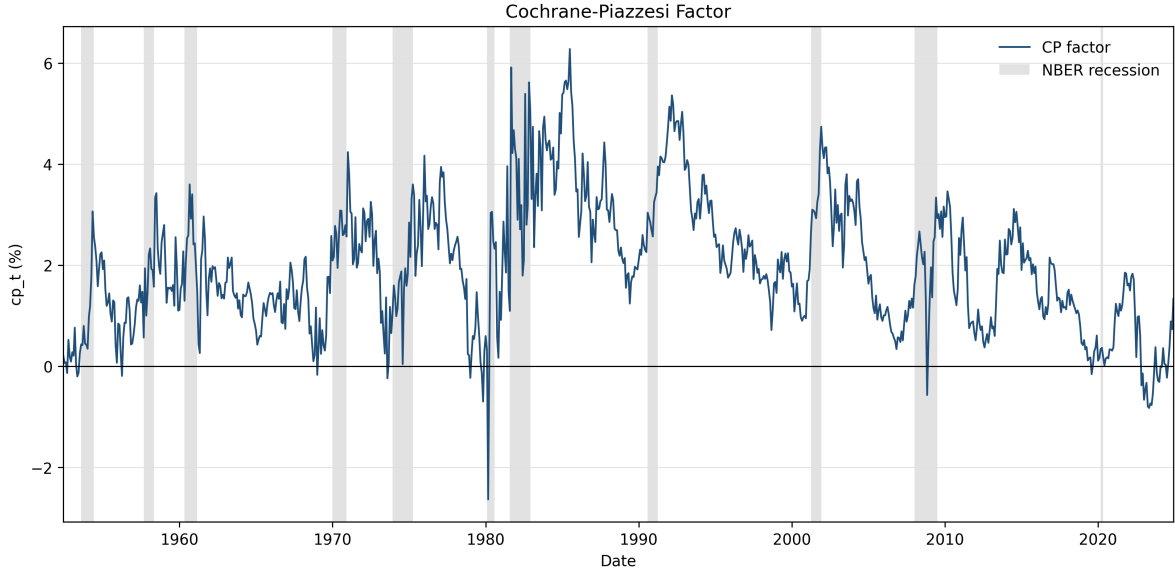


Figure 22: Cochrane–Piazzesi factor with NBER recession periods.

4.5 Part (e)

For $H = 2, 3, 4, 5$, I estimate the following predictive regression by OLS:

$$xr_{b,t+1}^{(H)} = a^{(H)} + b^{(H)}cp_t + \epsilon_t^{(H)},$$

where $xr_{b,t+1}^{(H)}$ is the one-year-ahead excess log return on an H -year bond and cp_t is the Cochrane–Piazzesi (CP) factor constructed in Q4(d) as a linear combination of the five current forward rates. I use Newey–West standard errors with automatic lag selection to account for heteroskedasticity and autocorrelation.

All four estimated coefficients are positive and statistically significant at the 1% level, indicating that higher values of the CP factor predict higher one-year-ahead excess bond returns across all four maturities. The estimated coefficients increase with maturity, from 0.4418 for two-year bonds to 1.4791 for five-year bonds, indicating greater sensitivity to the CP factor for longer-maturity bonds.

The R^2 values range from 13.67% to 17.05%, with the highest explanatory power for four-year bonds. Compared with Q4(c), the CP factor produces higher R^2 values at every maturity. This suggests that combining information from the entire forward-rate curve captures more variation in future excess returns than individual forward spreads.

Table 10: One-year excess bond return regressions on the CP factor

	2 years	3 years	4 years	5 years
<i>CP factor</i>				
$\widehat{b}^{(H)}$ [NW t -stat]	0.4418 [4.1257]	0.8274 [4.1243]	1.2517 [4.3896]	1.4791 [4.1986]
R^2	13.6703%	14.3774%	17.0468%	15.5228%